

## Letter of Motivation

### Introduction

My name is **Swathi, Ramesh** and I wish to apply for the **Master of Science in Computer Science** at **Technische Universität Berlin**. Here I envision gaining a substantial understanding of enhanced computing systems and acquiring suitable skills to succeed in the growing technological environment.

### Academic, Professional Experience, Technical Certifications, and Technical Skills

I hold a Bachelor of Engineering in **Computer Science and Engineering** from Visvesvaraya Technological University, India, with a **CGPA of 7.71 (German Grade: 2.3)**. I have a strong grounding in such courses as C Programming for Problem Solving, Data Structures and Applications, Computer Network and Security, Database Management System and Digital Electronics. Prior to this, I completed my Pre-University (12th Class) with a score of **78% (German Grade: 2.0)** and my 10th class exam mark sheet from the Indian Certificate of Secondary Education with **87.84 %** passing marks. I have gained practical experience through several internships. At **Varcons Technologies Pvt Ltd.**, I worked on **Twitter Sentiment Analysis** during the COVID-19 pandemic, where I applied **machine learning models** to analyze real-time data and presented a comprehensive report on the findings. Additionally, I interned at **Karnataka German Technical Training Institute**, where I completed a **Red Hat Certified Engineer (RHCE)** program, equipping me with valuable skills in Linux systems and network administration. My **Bachelor's project**, titled "**Design and Fabrication of Smart Nursing Robot**," aimed at addressing the growing demand for **smart healthcare systems** by leveraging **AI and machine learning** technologies to assist elderly and disabled individuals with personalized healthcare services. In terms of **certifications**, I have successfully completed the **Java Full Stack course** from **Pentagon Space**, a course on **IoT Edge Computing and IoT Analytics** from **Infosys**, and a training program in **Red Hat (RHCE)** from **Karnataka German Technical Training Institute Bengaluru**. These experiences, combined with my **technical skills** in **Network Analysis and Synthesis, Industrial Automation, Digital Electronics, Telecom**, and familiarity with **IoT and machine learning**, have significantly enhanced my competencies and deepened my motivation to pursue advanced studies in **Artificial Intelligence**. Also, I Completed 16 years of Education in the English Language.

### Why Master of Science in Computer Science?

The objectives of the proposed **Master of Science in Computer Science** at **TU Berlin** are well-defined. Its curriculum provides a sufficient learning framework that corresponds to my academic profile and career plan. Therefore, I believe the program's emphasis on algorithms, machine learning, and system architecture will help develop enhanced knowledge about intricate computational structures. Considering the unceasing demand for specialists focusing on AI, IoT, and data science, this course will help me shape the market leader. This will also improve my problem-solving skills because the program is theoretically interrelated and blended with practical practices. For this same reason, I believe that the focus of the course on conducting independent research on emerging technologies is well suited to allow me to help advance the technology field. Also, I am highly interested in sharing ideas with students and tutors on the practical tasks of initiating innovative projects in the field. I anticipate that the successful finish of this course information and communication technology will have prepared me into an all-round computer scientist with marketable opportunities for work in other countries within the industry and the universities.

### Why Germany?

The German education sector has developed a great reputation for offering quality education, especially in engineering and technology. The country has a tremendously advanced research and innovation agenda which makes it a perfect ground to pursue a master's degree in computer science. **Germany** hosts some of the leading tech firms and accelerating as well as innovative firms which make it a good location for students to learn more about the firms within the industry. However, **Germany** boasts of a healthy economy and an ever-shifting focus on digitalization meaning the jobs will continue to be in demand and available for qualified tech workers. Students from other countries such as me may be able to acquire quality education since most of the courses taught are in English. Besides, Germans have a diversified historical background, and the country is friendly to international students.

### Why TU Berlin?

**Technische Universität Berlin** is a top European university that offers education and conducts research in computer science. It has equipped all its host facilities with modern infrastructures and offers a unique research Experience for its learners. I am especially surprised that Computer Science is very involved with fields that interest me like AI, IoT, and cybersecurity. The university has a great location in Berlin, being one of the most popular places for technology startups and IT giants to expand. **TU Berlin's** focus on international cooperation will also enable me to collaborate with varied groups of people and obtain worldwide views of technological issues. Further, reliable production with industry companies ensures that I easily get employment immediately after the university.

### Conclusion

Lastly, I believe that studying at **TU Berlin** and enrolling in the **Master of Science in Computer Science** program will help me accomplish my objective. I am looking forward to being able to partake in as well as being able to gain from the active academic and research environment at **TU Berlin**.

**Thank you for considering my application.**

**Sincerely,**

**Swathi, Ramesh**